

LeetCode 1043: Max Sum After Partitioning

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Key Notes & Explanations

1. Range Max Precomputation:

$dp[l][r]$ is used to find the maximum in that range "so that baar baar calculate na karu".

2. Partition Condition (NEED TO READ CAREFULLY):

"k partition nhi , atmost k length ki partitions allowed he".

3. State Definition:

"so need only $dp[i]$ not $dp[i][k]$ ".

$dp[i]$ will give the max sum after placing i elements.

4. Transitions:

Using $\max(0, i-k)$ — "kyunki k length ki hi partition ho skti last waali bhi".

C++ Source Code

```
class Solution {
public:
    int maxSumAfterPartitioning(vector<int>& arr, int k) {
        int n=arr.size();

        vector<vector<int>>> rangemax(n+1,vector<int>(n+1,INT_MIN));
        //dp[l][r] to find max in that range so that baar baar calculate na karu

        for(int l=0;l<n;l++){
            int maxuuu=INT_MIN;
            for(int r=l;r<n;r++){
                maxuuu=max(maxuuu,arr[r]);
                rangemax[l][r]=maxuuu;
            }
        }

        //NEED TO READ CAREFULLY
        //k partition nhi , atmost k length ki partitions allowed he
```

```

//so need only dp[i] not dp[i][k]
//dp[i] will give max sum after placing i elements

vector<int> dp(n+1,INT_MIN);
dp[0]=0;

for(int i=1;i<=n;i++){
    for(int j=i-1;j>=max(0,i-k);j--){
        //max(0,i-k) kyunki k length ki hi partition ho skti last waali bhi

        int maxu=rangemax[j][i-1];
        int curr=dp[j]+(maxu*(i-j));

        dp[i]=max(dp[i],curr);
    }
}

return dp[n];
};

```